

Assignment 4: Using GenAI to Prepare for the Midterm

Module 4 | Scottie Dunnavaan

Acknowledgment of GenAI Use

GenAI Tool Used: Gemini 1.5 Pro

Purpose: Generating test essay questions, critiquing initial draft answers, and performing gap-finding verification on updated answers for Assignment 4 midterm preparation.

Part 1: Generated Essay Questions (15 Points)

PROMPT ENTERED

Act as a computer science professor for a machine learning course. Generate 3 basic short-answer essay questions testing introductory concepts on Deep Neural Networks (DNNs), Loss Functions, and Convolutional Neural Networks (CNNs). Keep the questions simple and direct.

Generated Questions:

- 1. Deep Neural Networks:** What is the main purpose of an activation function in a neural network, and why can't we just use linear functions everywhere?
- 2. Loss Functions:** What is the primary role of a loss function in training a neural network, and how does gradient descent use it?
- 3. Convolutional Neural Networks:** How does a convolutional layer process an input image, and why is it better for images than a regular dense (fully connected) layer?

Part 2: Initial Answers (20 Points)

Answer 1 (Deep Neural Networks)

An activation function decides whether a neuron should fire or not based on its input. We need non-linear activation functions like ReLU or Sigmoid because if we only used linear functions, the whole network would just behave like a single linear regression model, no matter how many hidden layers we add.

Answer 2 (Loss Functions)

A loss function measures the difference between the model's predictions and the actual target values to give a single error score. Gradient descent uses this score to calculate the slope (gradient) and shifts the model's weights in the opposite direction to make the error smaller over time.

Answer 3 (Convolutional Neural Networks)

A convolutional layer slides a small grid of numbers called a filter or kernel across the image to multiply against pixel values. This extracts key visual features like edges and textures into a feature map. It works better than a dense layer because it uses fewer parameters by sharing weights across the whole image.

Part 3: GenAI Critique (20 Points)

PROMPT ENTERED

Please critique my initial answers to the 3 essay questions below. Identify any technical inaccuracies, imprecise terminology, or missing concepts in each answer:

Question 1: What is the main purpose of an activation function in a neural network, and why can't we just use linear functions everywhere?

Answer 1: An activation function decides whether a neuron should fire or not based on its input. We need non-linear activation functions like ReLU or Sigmoid because if we only used linear functions, the whole network would just behave like a single linear regression model, no matter how many hidden layers we add.

Question 2: What is the primary role of a loss function in training a neural network, and how does gradient descent use it?

Answer 2: A loss function measures the difference between the model's predictions and the actual target values to give a single error score. Gradient descent uses this score to calculate the slope (gradient) and shifts the model's weights in the opposite direction to make the error smaller over time.

Question 3: How does a convolutional layer process an input image, and why is it better for images than a regular dense (fully connected) layer?

Answer 3: A convolutional layer slides a small grid of numbers called a filter or kernel across the image to multiply against pixel values. This extracts key visual features like edges and textures into a feature map. It works better than a dense layer because it uses fewer parameters by sharing weights across the whole image.

GENAI RESPONSE

Critique on Answer 1:

Accurate: Stating that stacking linear functions collapses the deep network into a single linear model.

Inaccurate/Imprecise: Describing an activation function as deciding "whether a neuron fires" uses a biological step-function analogy. Modern continuous activations (like ReLU or Sigmoid) scale continuous outputs rather than outputting binary firing states.

Missing Concept: Needs to explicitly state that non-linearity enables the network to learn complex non-linear decision boundaries and approximate non-linear functions.

Critique on Answer 2:

Accurate: Identifying the loss function as producing a scalar error metric and noting that parameters update in the opposite direction of the gradient.

Inaccurate/Imprecise: Saying "gradient descent uses this score to calculate the slope" is technically flawed. Gradient descent relies on backpropagation (the calculus chain rule) to compute partial derivatives of the loss function with respect to each weight.

Missing Concept: Omitted the learning rate, which scales the magnitude of the parameter update step.

Critique on Answer 3:

Accurate: Explaining the sliding kernel mechanism and weight sharing parameter reduction.

Inaccurate/Imprecise: "Multiply against pixel values" is incomplete—the operation is an element-wise multiplication followed by a summation (a dot product) at each spatial window.

Missing Concept: Failed to mention spatial locality (preserving 2D grid structure) and translation equivariance. Dense layers flatten images into 1D vectors, destroying spatial relationships and causing a parameter explosion.

Part 4: Updated Answers, Gap Finding, and Changes (20 Points)

Updated Refined Answers (~100% Accurate, Concise)

Updated Answer 1 (Deep Neural Networks)

An activation function introduces non-linearity into a neural network, enabling it to model complex, non-linear patterns. Modern continuous activations scale inputs non-linearly rather than making binary firing decisions. Without non-linear activations, stacking multiple layers mathematically collapses into a single linear transformation, making depth useless.

Updated Answer 2 (Loss Functions)

A loss function quantifies the discrepancy between model predictions and true targets into a scalar error value. Gradient descent uses backpropagation (the calculus chain rule) to compute the partial derivative of this loss with respect to each weight, updating parameters in the opposite direction scaled by a learning rate to minimize overall error.

Updated Answer 3 (Convolutional Neural Networks)

A convolutional layer slides a learnable kernel across an image, computing element-wise multiplications and sums at each position to form a feature map. It outperforms dense layers because it preserves 2D spatial locality, leverages translation equivariance, and drastically reduces parameters through weight sharing instead of flattening the image into a 1D vector.

PROMPT ENTERED

Please review my updated answers to the 3 essay questions. Are there any remaining technical gaps, errors, or missing key points, or are these answers complete and accurate?

Updated Answer 1: An activation function introduces non-linearity into a neural network, enabling it to model complex, non-linear patterns. Modern continuous activations scale inputs non-linearly rather than making binary firing decisions. Without non-linear activations, stacking multiple layers mathematically collapses into a single linear transformation, making depth useless.

Updated Answer 2: A loss function quantifies the discrepancy between model predictions and true targets into a scalar error value. Gradient descent uses backpropagation (the calculus chain rule) to compute the partial derivative of this loss with respect to each weight, updating parameters in the opposite direction scaled by a learning rate to minimize overall error.

Updated Answer 3: A convolutional layer slides a learnable kernel across an image, computing element-wise multiplications and sums at each position to form a feature map. It outperforms dense layers because it preserves 2D spatial locality, leverages translation equivariance, and drastically reduces parameters through weight sharing instead of flattening the image into a 1D vector.

GENAI RESPONSE

Your updated answers are accurate, technically precise, and concise.

Question 1: Successfully corrected the biological 'firing' metaphor to continuous non-linear mapping and correctly noted mathematical linear collapse.

Question 2: Correctly incorporated backpropagation, partial derivatives with respect to individual parameters, and the scaling role of the learning rate.

Question 3: Accurately specified the element-wise sum-product operation, noted 2D spatial locality, translation equivariance, and contrast with 1D image flattening in dense layers.

No critical technical gaps remain for an introductory midterm level.

Explanation of What Changed and Why

Question 1

What changed: Replaced the "neuron firing" phrasing with explicit mention of non-linear mathematical mappings and scaling.

Why: Biological threshold analogies are technically inaccurate for modern continuous activation functions like ReLU or Sigmoid.

Question 2

What changed: Added backpropagation (chain rule), partial derivatives, and the learning rate.

Why: Gradient descent cannot compute parameter updates directly from the loss score alone; it requires partial derivatives calculated per weight via backpropagation and scaled by the step size.

Question 3

What changed: Clarified the kernel operation as “element-wise multiplication and summation” and added 2D spatial locality, translation equivariance, and 1D flattening drawbacks.

Why: Simple multiplication fails to describe 2D spatial cross-correlation, and omitting spatial locality misses the core advantage CNNs have over dense layers.

Part 5: Reflection & Recommendations (20 Points)

Written independently, without GenAI assistance, per assignment instructions.

Using Generative AI as an iterative tutor highlighted the gap between high-level intuition and technical precision. My initial responses contained common conceptual shortcuts, such as confusing activation functions with binary firing thresholds or omitting backpropagation when describing gradient updates. Prompting the AI to critique my answers exposed these specific technical gaps without providing unearned shortcuts. Resubmitting revised definitions enforced active recall and precise mathematical vocabulary.

To effectively learn using Generative AI, I recommend a strict two-stage workflow: first, draft responses independently to establish a baseline of personal understanding, and second, prompt the AI specifically as an evaluator rather than a generator. Ask targeted questions like “Identify missing technical terms or mathematical inaccuracies in my explanation” rather than “Write the answer for me.” This forces active engagement, prevents over-reliance on generated text, and builds deep conceptual mastery required for exams.

Part 6: Overall Compliance (5 Points)

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Acknowledgment: Generative AI (Gemini 1.5 Pro) was utilized for question generation, answer critique, and gap verification. All initial answers, revised explanations, and reflections reflect personal work and verification.